Inputs and outputs in detail

Model inputs

29 × 10 history features + 29 masks

22 derived static features

108 taxon tokens + unknown token

At least 8 prior fixes spanning ≥6h

Model outputs

9 horizons: 1h, 12h, 1, 3, 7, 14, 30, 60, 90d

Samples from a joint route distribution

Location and uncertainty at each horizon

No learned animal-ID embedding

Local RTX 4060 Laptop: 108 tracks × 64 futures × 9 horizons in ≈41 ms

Warm model and prepared inputs; coordinate conversion included; raw-track preparation and loading excluded.

Sources: migration_repaired_features.py; packaged configs; inference benchmark; data manifest.

Dataset split and coverage

TRAIN	**799,681**	Cutoff through 2020
Validation: unseen animals	**201,563**	Conservative identity exclusions
Validation: held-out year	**43,896**	2021
Reserved assessment	**204,266**	Targets remain sealed

Marine examples: green and hawksbill turtles, grey seals, sperm whales and seabirds.

144 study labels have cases; 131 have TRAIN cases. Coverage is uneven and is not a sample of all animal migration.

Sources: corrected cache manifest, preflight, study-inventory.csv and taxon-inventory.csv.

Architecture and evaluation details

Predictor	Width / layers	Ensemble share
Direct compact	128 / 2	**8.6%**
Direct scaled	512 / 6	**45.6%**
Goal route compact	128 / 2	**14.2%**
Goal route scaled	512 / 6	**31.6%**

TRAIN-only normalization; square-root taxon sampling; NLL + energy optimization.

Full development: 245,459 origins, 1,734,789 observed targets and 64 draws at nine horizons.

Sources: packaged checkpoint configurations; migration_train.py; migration_evaluate.py; migration_ensemble.py.

What the score means and what it proves

Energy score

Rewards samples near the observed future, with a correction for forecast dispersion. Lower is better; units are kilometres.

Mean point error

Measures the miss of the forecast mean. It answers a different question from the full distribution score.

What we can claim

Better energy on the tested local panel

What still needs evidence

Independent transfer, official benchmark ranking and global frontier superiority

Evidence: exact local report hashes in evidence.json. Method code: github.com/cifkao/moveformer; github.com/swforrest/deepSSF_package.